



```
In [1]: import pandas as pd
```

```
In [2]: employees = pd.DataFrame({
    'employee_id': [1, 2, 3, 4, 5],
    'name': ['John', 'Anna', 'Peter', 'Linda', 'Bob'],
    'department': ['HR', 'IT', 'Finance', 'IT', 'HR']
})

# DataFrame 2: Salary information
salaries = pd.DataFrame({
    'employee_id': [1, 2, 3, 6, 7],
    'salary': [60000, 80000, 65000, 70000, 90000],
    'bonus': [5000, 10000, 7000, 8000, 12000]
})
```

```
In [3]: employees
```

```
Out[3]:
```

	employee_id	name	department
0	1	John	HR
1	2	Anna	IT
2	3	Peter	Finance
3	4	Linda	IT
4	5	Bob	HR

```
In [4]: salaries
```

```
Out[4]:
```

	employee_id	salary	bonus
0	1	60000	5000
1	2	80000	10000
2	3	65000	7000
3	6	70000	8000
4	7	90000	12000

```
In [6]: pd.merge(employees, salaries, on="employee_id", how= "inner")
```

```
Out[6]:
```

	employee_id	name	department	salary	bonus
0	1	John	HR	60000	5000
1	2	Anna	IT	80000	10000
2	3	Peter	Finance	65000	7000

```
In [7]: pd.merge(employees, salaries, on="employee_id", how= "outer")
```

Out[7]:

	employee_id	name	department	salary	bonus
0	1	John	HR	60000.0	5000.0
1	2	Anna	IT	80000.0	10000.0
2	3	Peter	Finance	65000.0	7000.0
3	4	Linda	IT	NaN	NaN
4	5	Bob	HR	NaN	NaN
5	6	NaN	NaN	70000.0	8000.0
6	7	NaN	NaN	90000.0	12000.0

In [8]: `pd.merge(employees, salaries, on="employee_id", how= "left")`

Out[8]:

	employee_id	name	department	salary	bonus
0	1	John	HR	60000.0	5000.0
1	2	Anna	IT	80000.0	10000.0
2	3	Peter	Finance	65000.0	7000.0
3	4	Linda	IT	NaN	NaN
4	5	Bob	HR	NaN	NaN

In [9]: `pd.merge(employees, salaries, on="employee_id", how= "right")`

Out[9]:

	employee_id	name	department	salary	bonus
0	1	John	HR	60000	5000
1	2	Anna	IT	80000	10000
2	3	Peter	Finance	65000	7000
3	6	NaN	NaN	70000	8000
4	7	NaN	NaN	90000	12000

In [10]:

```
df1 = pd.DataFrame({
    'A': ['A0', 'A1', 'A2'],
    'B': ['B0', 'B1', 'B2'],
    'C': ['C0', 'C1', 'C2']
})

df2 = pd.DataFrame({
    'A': ['A3', 'A4', 'A5'],
    'B': ['B3', 'B4', 'B5'],
    'C': ['C3', 'C4', 'C5']
})
```

```
In [11]: df1
```

```
Out[11]:
```

	A	B	C
0	A0	B0	C0
1	A1	B1	C1
2	A2	B2	C2

```
In [12]: df2
```

```
Out[12]:
```

	A	B	C
0	A3	B3	C3
1	A4	B4	C4
2	A5	B5	C5

```
In [15]: pd.concat([df1,df2], axis= 1)
```

```
Out[15]:
```

	A	B	C	A	B	C
0	A0	B0	C0	A3	B3	C3
1	A1	B1	C1	A4	B4	C4
2	A2	B2	C2	A5	B5	C5

```
In [16]: df1 = pd.DataFrame({
    'name': ['Alice', 'Bob', 'Charlie']
}, index=[1, 2, 3])

# Second DataFrame
df2 = pd.DataFrame({
    'score': [85, 90, 75]
}, index=[2, 3, 4])
```

```
In [17]: df1
```

```
Out[17]:
```

	name
1	Alice
2	Bob
3	Charlie

```
In [18]: df2
```

Out[18]:

	score
2	85
3	90
4	75

```
In [22]: df1.join(df2, how="inner")
```

Out[22]:

	name	score
2	Bob	85
3	Charlie	90

```
In [23]: df1.join(df2, how="outer")
```

Out[23]:

	name	score
1	Alice	NaN
2	Bob	85.0
3	Charlie	90.0
4	NaN	75.0

```
In [26]: df3=df1.join(df2, how="right")
```

```
In [25]: df1.join(df2, how="left")
```

Out[25]:

	name	score
1	Alice	NaN
2	Bob	85.0
3	Charlie	90.0

```
In [27]: df3
```

Out[27]:

	name	score
2	Bob	85
3	Charlie	90
4	NaN	75

```
In [28]: df3.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 3 entries, 2 to 4
Data columns (total 2 columns):
#   Column  Non-Null Count  Dtype
---  -
0   name    2 non-null         object
1   score   3 non-null         int64
dtypes: int64(1), object(1)
memory usage: 72.0+ bytes
```

```
In [30]: df3["score"]=df3["score"].astype("str")
```

```
In [31]: df3
```

```
Out[31]:
```

	name	score
2	Bob	85
3	Charlie	90
4	NaN	75

```
In [32]: df3.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 3 entries, 2 to 4
Data columns (total 2 columns):
#   Column  Non-Null Count  Dtype
---  -
0   name    2 non-null         object
1   score   3 non-null         object
dtypes: object(2)
memory usage: 72.0+ bytes
```

```
In [33]: df3["score"]= df3["score"].astype("int")
```

```
In [34]: df3.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 3 entries, 2 to 4
Data columns (total 2 columns):
#   Column  Non-Null Count  Dtype
---  -
0   name    2 non-null         object
1   score   3 non-null         int64
dtypes: int64(1), object(1)
memory usage: 72.0+ bytes
```

```
In [42]: df2 = pd.DataFrame({
        'score': ["py85", "90", "75"]
    }, index=[2, 3, 4])
```

```
In [43]: df2.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 3 entries, 2 to 4
Data columns (total 1 columns):
#   Column  Non-Null Count  Dtype
---  -
0    score    3 non-null      object
dtypes: object(1)
memory usage: 48.0+ bytes
```

```
In [44]: df2
```

```
Out[44]:
```

	score
2	py85
3	90
4	75

```
In [45]: # df2["score"].astype("int")
```

```
In [47]: df2["score"] = df2["score"].str.replace("py", "")
```

```
In [48]: df2["score"].astype("int")
```

```
Out[48]:
```

2	85
3	90
4	75

Name: score, dtype: int64

```
In [49]: data = {
    'Category': ['A', 'B', 'A', 'B', 'A', 'B', 'A', 'B'],
    'Store': ['S1', 'S1', 'S2', 'S2', 'S1', 'S2', 'S2', 'S1'],
    'Sales': [100, 200, 150, 250, 120, 180, 200, 300],
    'Quantity': [10, 15, 12, 18, 8, 20, 15, 25],
    'Date': pd.date_range('2023-01-01', periods=8)
}
df = pd.DataFrame(data)
```

```
In [50]: df
```

```
Out[50]:
```

	Category	Store	Sales	Quantity	Date
0	A	S1	100	10	2023-01-01
1	B	S1	200	15	2023-01-02
2	A	S2	150	12	2023-01-03
3	B	S2	250	18	2023-01-04
4	A	S1	120	8	2023-01-05
5	B	S2	180	20	2023-01-06
6	A	S2	200	15	2023-01-07
7	B	S1	300	25	2023-01-08

```
In [52]: df.to_csv("data.csv", index=False)
```

```
In [54]: df5=pd.read_excel("test2.xlsx")
```

```
In [55]: df5
```

Out[55]:

	Date	Product	Region	Sales	Units	Rep
0	2023-01-01	A	East	365	78	John
1	2023-01-02	B	West	568	64	Mary
2	2023-01-03	C	North	823	25	Bob
3	2023-01-04	D	South	348	75	Alice
4	2023-01-05	A	East	103	43	John
5	2023-01-06	B	West	690	66	Mary
6	2023-01-07	C	North	830	92	Bob
7	2023-01-08	D	South	649	63	Alice
8	2023-01-09	A	East	879	88	John
9	2023-01-10	B	West	297	31	Mary
10	2023-01-11	C	North	225	86	Bob
11	2023-01-12	D	South	424	35	Alice
12	2023-01-13	A	East	987	57	John
13	2023-01-14	B	West	690	67	Mary
14	2023-01-15	C	North	720	71	Bob
15	2023-01-16	D	South	106	19	Alice
16	2023-01-17	A	East	748	68	John
17	2023-01-18	B	West	685	89	Mary
18	2023-01-19	C	North	740	76	Bob
19	2023-01-20	D	South	508	70	Alice

```
In [57]: df5.Units.max()
```

```
Out[57]: np.int64(92)
```

```
In [58]: df5.Units.min()
```

```
Out[58]: np.int64(19)
```

```
In [59]: func= lambda x: x*10
```

```
In [62]: df5["Units"]=df5["Units"].apply(func)
```

```
In [63]: df5
```

Out[63]:

	Date	Product	Region	Sales	Units	Rep
0	2023-01-01	A	East	365	780	John
1	2023-01-02	B	West	568	640	Mary
2	2023-01-03	C	North	823	250	Bob
3	2023-01-04	D	South	348	750	Alice
4	2023-01-05	A	East	103	430	John
5	2023-01-06	B	West	690	660	Mary
6	2023-01-07	C	North	830	920	Bob
7	2023-01-08	D	South	649	630	Alice
8	2023-01-09	A	East	879	880	John
9	2023-01-10	B	West	297	310	Mary
10	2023-01-11	C	North	225	860	Bob
11	2023-01-12	D	South	424	350	Alice
12	2023-01-13	A	East	987	570	John
13	2023-01-14	B	West	690	670	Mary
14	2023-01-15	C	North	720	710	Bob
15	2023-01-16	D	South	106	190	Alice
16	2023-01-17	A	East	748	680	John
17	2023-01-18	B	West	685	890	Mary
18	2023-01-19	C	North	740	760	Bob
19	2023-01-20	D	South	508	700	Alice

```
In [64]: df5["Region"].unique()
```

```
Out[64]: array(['East', 'West', 'North', 'South'], dtype=object)
```

```
In [70]: df5.groupby(["Product", "Region"])["Sales", "Units"].mean()
```

Out[70]:

		Sales	Units
Product	Region		
A	East	616.4	668.0
B	West	586.0	634.0
C	North	667.6	700.0
D	South	407.0	524.0

In []: